

Requirements Specification & Traceability Matrix (RTM)

Problem Statement #42: Internal Microservice Catalog & Health Portal | **Student:** Subramani B M (PES1UG24CS473) | Section H

1. Functional Requirements (FR-001 to FR-005)

ID	Type	Description	Acceptance Criteria	Rationale
FR-001	Health Monitoring	Pings health endpoints at 30s intervals; computes rolling 24h availability.	Pass: Downtime logged & alert sent on 3 failed probes. Fail: Down service marked healthy.	Ensures reliability and instant incident detection.
FR-002	Catalog Management	Searchable catalog of all microservices (name, owner, repo, env, docs).	Pass: New service appears within 60s. Fail: Service missing or stale metadata.	Single source of truth for microservice discovery.
FR-003	Dependency Mapping	Interactive dependency graph showing upstream/downstream relationships.	Pass: New dependency reflected in <2 min; clickable nodes. Fail: Missing/false edges.	Identifies blast radius and critical paths.
FR-004	Alerting & Notifications	Dispatches real-time alerts via Slack, email, and webhooks upon outage.	Pass: On-call contacts notified within 60s. Fail: Missed/misdirected alerts.	Minimizes MTTA and MTTR for incident teams.
FR-005	API Documentation	Aggregates auto-generated OpenAPI / Swagger specs for each service.	Pass: Spec renders in <2 min with test sandbox. Fail: Incomplete/broken schema.	Reduces developer onboarding time.

2. Non-Functional Requirements (NFR-001 & NFR-002)

ID	Type	Description	Acceptance Criteria	Rationale
NFR-001	Performance	Graph renders up to 200 nodes smoothly with pan/zoom <2s.	Pass: Render ≤2s, interaction ≤200ms under 50 users. Fail: >2s latency.	Fluid tracing during outage triage.
NFR-002	Security	TLS 1.2+ encryption, SSO (OAuth2/SAML), and vault-backed secrets.	Pass: Zero plaintext credentials; strict RBAC. Fail: Plaintext leak or auth bypass.	Protects infrastructure topology & tokens.

3. Requirements Traceability Matrix (RTM)

Req ID	Description	Use Case	Component(s)	Test Method	Status
FR-001	30s Probing & 24h Availability	UC-004: Health Monitor	Health Monitoring, TimescaleDB	Automated synthetic pinger test	Verified
FR-002	Searchable Service Catalog	UC-001, UC-002: Catalog Search	Catalog Service, PostgreSQL	CRUD registration & indexing test	Verified
FR-003	200-Node Dependency Graph	UC-003: View Dep Graph	Dependency Mapping, Neo4j	200-node topology render benchmark	Verified
FR-004	Multi-Channel Alerts (<60s)	UC-004, UC-006: Outage Alert	RabbitMQ, Alert Service, Slack	Outage injection & dispatch timer	Verified
FR-005	Aggregated OpenAPI Docs	UC-005: Browse API Docs	API Doc Aggregator, Redis	OpenAPI 3.0 schema parsing test	Verified
NFR-001	Sub-2.0s Render SLA	UC-003: View Dep Graph	Dependency Mapping, Neo4j	JMeter 50 concurrent session test	Verified
NFR-002	SSO & Secrets Vault Isolation	All Use Cases (UC-001..006)	API Gateway, HashiCorp Vault	OWASP ZAP scan & mTLS check	Verified